Varukolu Vishal

cs23btech11064@iith.ac.in | vv14022006@gmail.com | +91 7780128500 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.00
Narayana junior college	Class XII (State board)	2023	96.10%
Narayana high school	Class X (State board)	2021	10.00%

SKILLS

Languages: C, C++, Python, HTML

Python Libraries: NumPy, Pandas, Matplotlib, PyTorch

Databases, Systems & Concepts: MySQL, MongoDB, Object-Oriented Programming (OOP), RISC-V

Developer Tools: Git, Vim, VS Code, Bash, Docker, LaTeX

Environments: Linux, Windows

PROJECTS

Analytics-BI - Business Intelligence Platform (Group Project) GitHub

- Implemented the backend data ingestion pipeline in Node.js/Express, streaming CSV/XLS/XLSX uploads into MongoDB GridFS and automatically inferring schemas while quarantining invalid rows into a Dead Letter Queue for review and restoration.
- Built Redis,BullMQ background workers for bulk ingestion and export jobs (CSV, XLSX, PDF, PNG), using Socket.IO to push real-time progress updates to the React frontend.
- Added backend security and collaboration features, including RBAC middleware, input sanitization/rate limiting, and draft/publish dashboard APIs with version-based conflict handling.

Multi-Agent Orchestration System GitHub

- Designed typed state schemas and custom reducers for a LangGraph-based multi-agent system, defining how agent state transitions and merges across specialized roles like router, researcher, and critic.
- Built a provider-neutral tool registry with JSON Schema validation and runtime adapters that translate tool definitions between OpenAI function calling and Anthropic tool use formats without rewriting tool code.
- Implemented a PostgreSQL persistence layer using SQLAlchemy 2.0 and Alembic, structuring a repository pattern across task, run, message, tool call, and HITL entities backed by 130+ unit tests.

JobPilot - AI-Powered Job Search Assistant GitHub

- Built a full-stack job search platform with Next.js and PostgreSQL that automates resume parsing, job discovery via the Adzuna API, and application tracking across multiple regions.
- Designed a GPT-4o pipeline that converts PDF resumes into structured candidate profiles and scores job postings (0-100) against user skills, highlighting matched qualifications and gaps.
- Integrated Browserbase and Stagehand to automate company research--navigating sites, extracting tech stack and culture data, and persisting structured reports in PostgreSQL via InsForge.

Self-Healing RAG -- GitHub

- Built a self-correcting RAG pipeline using LangGraph, ChromaDB, Sentence-Transformers, and Gemini/Hugging Face LLMs.
- Implemented Critic-driven loops for query reformulation, grounded regeneration, and safe abstention to reduce hallucinations.
- Benchmarked against baseline RAG with an evaluation harness and 129 passing pytest tests; supports offline and live modes.

LAOP: Linear Algebra DSL Compiler GitHub

- Built the compiler backend for a linear algebra DSL in C++, generating Eigen-based C++ code from parsed programs with support for scalars, vectors, matrices, control flow, and user-defined functions.
- Implemented semantic analysis, symbol table management, type checking, and matrix dimension validation to catch incorrect operations at compile time before code generation.
- Added optimization passes for matrix-chain multiplication ordering to reduce redundant computation and produce more efficient generated C++ code.

Neural Image Captioning with Xception + LSTM -- GitHub

- Built an end-to-end image captioning model using a pretrained Xception CNN encoder and LSTM decoder in TensorFlow/Keras on the Flickr8k dataset.
- Implemented cached feature extraction, tokenizer-based sequence framing, vectorized batch generation, and model checkpointing for efficient training.
- Added single-image inference and modernized the pipeline for TensorFlow 2.16+/Keras 3 with .keras and legacy .h5 support.

RELEVANT COURSES

Computer Science: Data Structures, Algorithms, Theory of Computation, Operating Systems , Database Management Systems , Computer Networks, Compilers , Software Engineering, Computer Architecture, Software Development Fundamentals, Topics in Theoretical Computer Science

AI/ML: Foundations of Machine Learning, Deep Learning, Multi-Agent Systems, Distributed AI Training , Linear Optimization

Mathematics: Discrete Mathematics, Linear Algebra, Probability & Random Variables, Introduction to Statistics, Calculus , Transform Techniques, Computational Number Theory & Algebra

EXTRACURRICULAR

LeetCode: Solved 500 Data Structures & Algorithms problems(profile)